

direction along the mRNA. Each codon specifies which one of the 20 amino acids will be incorporated at the corresponding position along a polypeptide. Because codons are nucleotide triplets, the number of nucleotides making up a genetic message must be three times the number of amino acids in the protein product. For example, 300 nucleotides along an mRNA strand code for 100 amino acids in the polypeptide it encodes.

Cracking the Code

Molecular biologists cracked the genetic code of life in the early 1960s when a series of elegant experiments disclosed the amino acid translations of each of the mRNA codons. The first codon was deciphered in 1961 by Marshall Nirenberg, of the National Institutes of Health, and colleagues. They synthesized an artificial mRNA made up of only uracil-bearing nucleotides, thus containing only one codon (UUU) over and over. When added to a test-tube mixture that contained all 20 amino acids, ribosomes, and the other components required for protein synthesis, the “poly-U” mRNA was translated into a polypeptide containing many units of the phenylalanine (Phe, or F) amino acids, strung together as a polypeptide chain. Thus, Nirenberg determined that the mRNA codon UUU specifies the amino acid phenylalanine. Soon, the amino acids specified by the codons AAA, GGG, and CCC were similarly identified.

Although more elaborate techniques were required to decode mixed triplets such as AUA and CGA, all 64 codons were deciphered by the mid-1960s. As **Figure 17.6** shows, 61 of the 64 triplets code for amino acids. The other three codons act as “stop” signals, or termination codons, marking the end of translation. Notice that the codon AUG has a dual function: It codes for the amino acid methionine (Met, or M) and also functions as a “start” signal, or initiation codon. Genetic messages usually begin with the mRNA codon AUG, which signals the protein-synthesizing machinery to begin translating the mRNA at that location. (Because AUG also stands for methionine, polypeptide chains all begin with methionine when they are synthesized. However, an enzyme may subsequently remove this starter amino acid from the chain.)

Notice in **Figure 17.6** that there is redundancy in the genetic code, but no ambiguity. For example, although codons GAA and GAG both specify glutamic acid (redundancy), neither of them ever specifies any other amino acid (no ambiguity). The redundancy in the code is not altogether random. In many cases, codons that are synonyms for a particular amino acid differ only in the third nucleotide base of the triplet. We will see why this is later in the chapter.

Our ability to extract the intended message from a written language depends on reading the symbols in the correct groupings—that is, in the correct **reading frame**. Consider this statement: “The red dog ate the bug.” Group the letters incorrectly by starting at the wrong point, and the result

▼ **Figure 17.6 The codon table for mRNA.** The three nucleotide bases of an mRNA codon are designated here as the first, second, and third bases, reading in the 5′ → 3′ direction along the mRNA. The codon AUG not only stands for the amino acid methionine (Met, or M) but also functions as a “start” signal for ribosomes to begin translating the mRNA at that point. Three of the 64 codons function as “stop” signals, marking where ribosomes end translation. Both one- and three-letter codes are shown for the amino acids; see **Figure 5.14** for their full names.

		Second mRNA base				
		U	C	A	G	
First mRNA base (5' end of codon)	U	UUU } Phe (F) UUC } UUA } Leu (L) UUG }	UCU } UCC } Ser (S) UCA } UCG }	UAU } Tyr (Y) UAC } UAA Stop UAG Stop	UGU } Cys (C) UGC } UGA Stop UGG Trp (W)	U C A G
	C	CUU } CUC } Leu (L) CUA } CUG }	CCU } CCC } Pro (P) CCA } CCG }	CAU } His (H) CAC } CAA } Gln (Q) CAG }	CGU } CGC } Arg (R) CGA } CGG }	U C A G
	A	AUU } Ile (I) AUC } AUA } AUG Met (M) or start	ACU } ACC } Thr (T) ACA } ACG }	AAU } Asn (N) AAC } AAA } Lys (K) AAG }	AGU } Ser (S) AGC } AGA } Arg (R) AGG }	U C A G
	G	GUU } GUC } Val (V) GUA } GUG }	GCU } GCC } Ala (A) GCA } GCG }	GAU } Asp (D) GAC } GAA } Glu (E) GAG }	GGU } GGC } Gly (G) GGA } GGG }	U C A G
						Third mRNA base (3' end of codon)

VISUAL SKILLS A segment in the middle of an mRNA has the sequence 5′-AGAGAACCGCGA-3′. Using the codon table, translate this sequence, assuming the first three nucleotides are a codon.

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will probably be gibberish: for example, “her edd oga tet heb ug.” The reading frame is also important in the molecular language of cells. The short stretch of polypeptide shown in **Figure 17.5**, for instance, will be made correctly only if the mRNA nucleotides are read from left to right (5′ → 3′) in the groups of three shown in the figure: UGG UUU GGC UCA. Although a genetic message is written with no spaces between the codons, the cell’s protein-synthesizing machinery reads the message as a series of nonoverlapping three-letter words. The message is *not* read as a series of overlapping words—UGGUUU, and so on—which would convey a very different message.

Evolution of the Genetic Code

EVOLUTION The genetic code is nearly universal, shared by organisms from the simplest bacteria to the most complex plants and animals. The mRNA codon CCG, for instance, is translated as the amino acid proline in all organisms whose